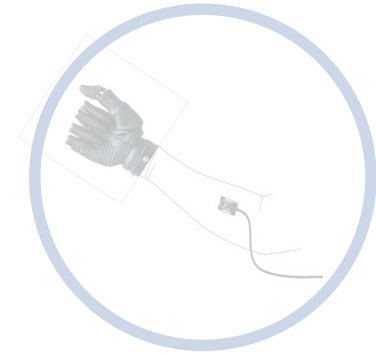
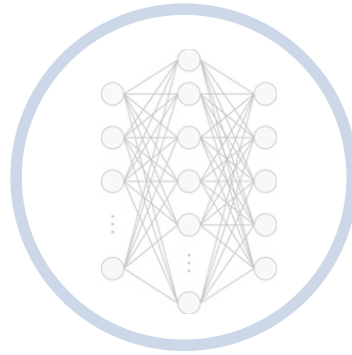
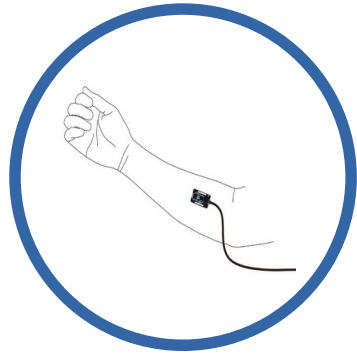


AI-Based Control System for Hand Gesture Prediction of RYO Smart Prosthetic Hand

Barret Feusi

Primary Objective



sEMG



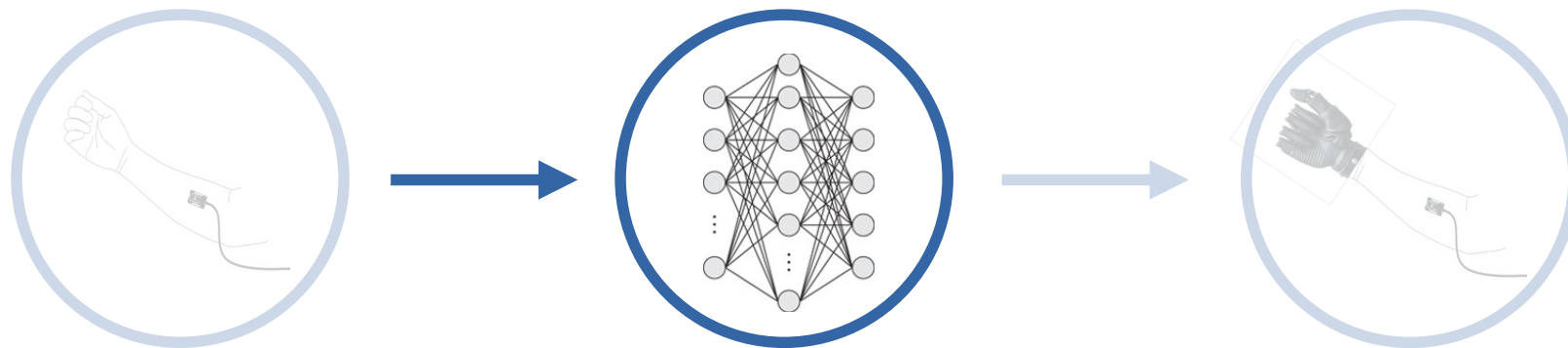
FMG



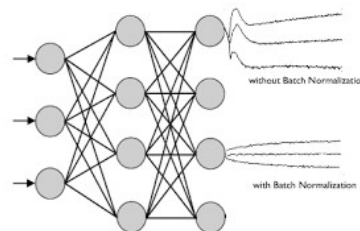
ESP32-C3 Supermini

To make a “real-time [biosignal]
acquisition and processing module...”

Primary Objective

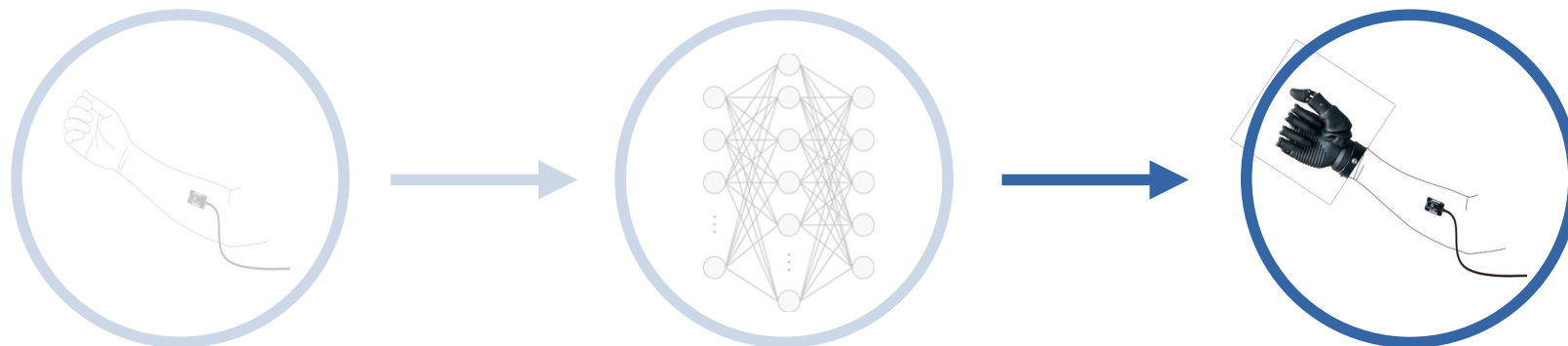


...with robust filtering

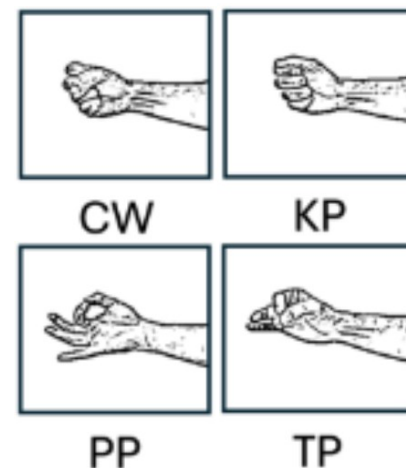


and normalization”...

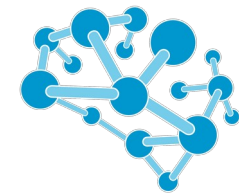
Primary Objective



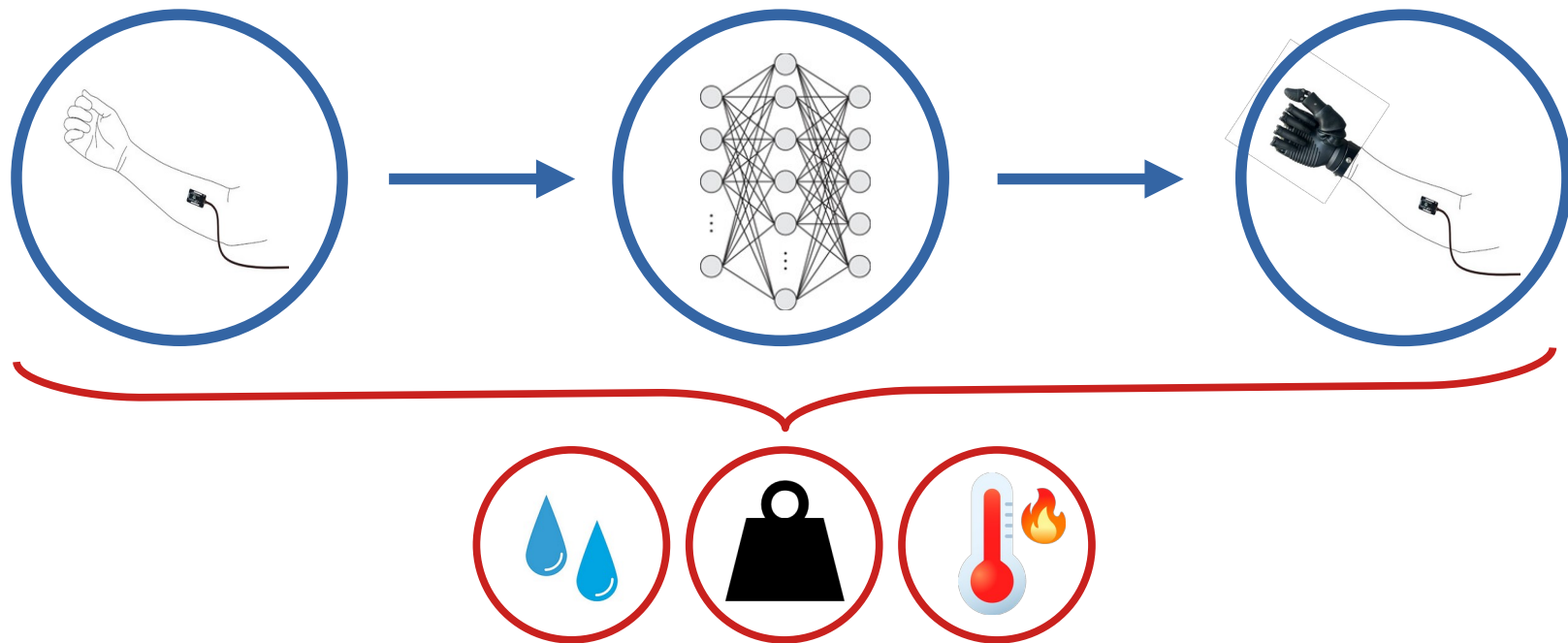
...to predict one of four hand gestures
for controlling an **affordable**
smart hand prosthesis.



Primary Objective

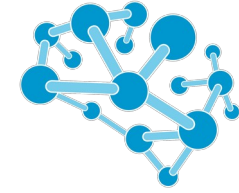


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AND have it work under non-ideal conditions!

Specific Objectives

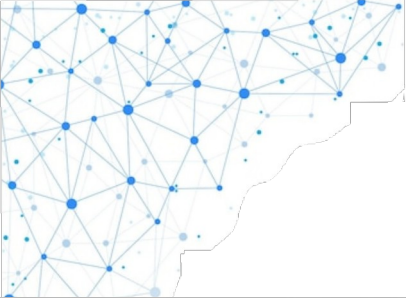


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- Train AI model on sEMG (and FMG) biosignals in ideal and non-ideal conditions. 
- Fit trained model on mini microcontroller with WiFi capabilities. 
- Develop app platform for calibrating model and sending real-time feedback and managing performance metrics (for personalized adaptation). 

Justification

“Today, daily life is composed of many computing systems, therefore interacting with them in a natural way makes the communication process more comfortable.... One form of [Human-Computer Interface] is Hand Gesture Recognition (HGR), which predicts the class and the instant of execution of a given movement of the hand. One possible input for these models is surface electromyography (EMG), which records the electrical activity of skeletal muscles. EMG signals contain information about the intention of movement generated by the human brain.”¹



User Profile



Science & Technology

Hand in bionic hand: The new age of accessible, high-tech prosthetics



RESEARCH ARTICLE

The effects of limb position and grasped load on hand gesture classification using electromyography, force myography, and their combination

Peyton R. Young¹, Kihun Hong¹, Eden J. Winslow², Giancarlo K. Sagastume³, Marcus A. Batraw⁴, Richard S. Whittle¹, Jonathon S. Schofield^{1*}

1 Department of Mechanical and Aerospace Engineering, University of California—Davis, Davis, California, United States of America, 2 Department of Biomedical Engineering, University of California—Davis, Davis, California, United States of America, 3 Department of Electrical and Computer Engineering, University of California—Davis, Davis, California, United States of America, 4 Department of Mechanical and Mechatronic Engineering and Advanced Manufacturing, California State University, Chico, Chico, California, United States of America

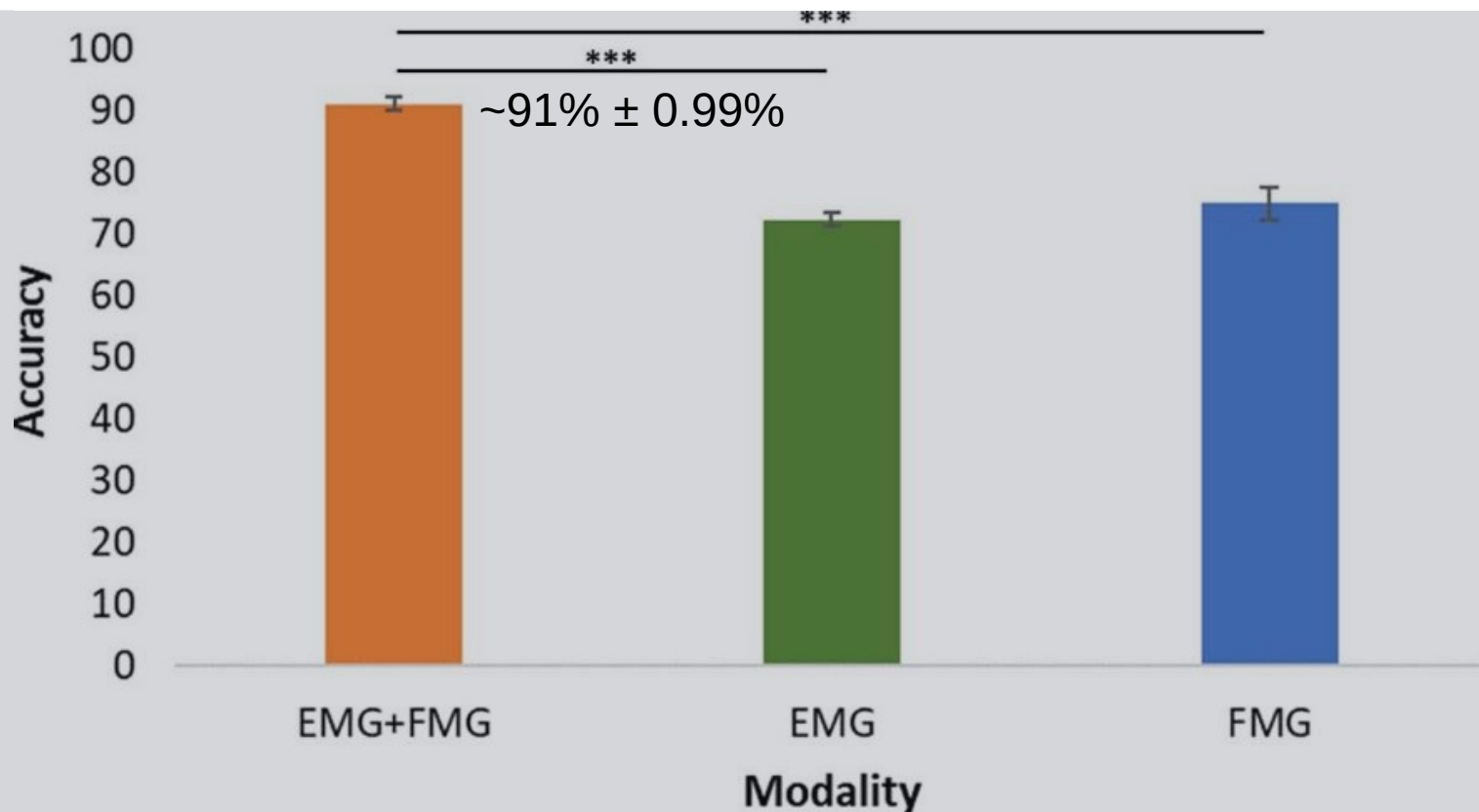
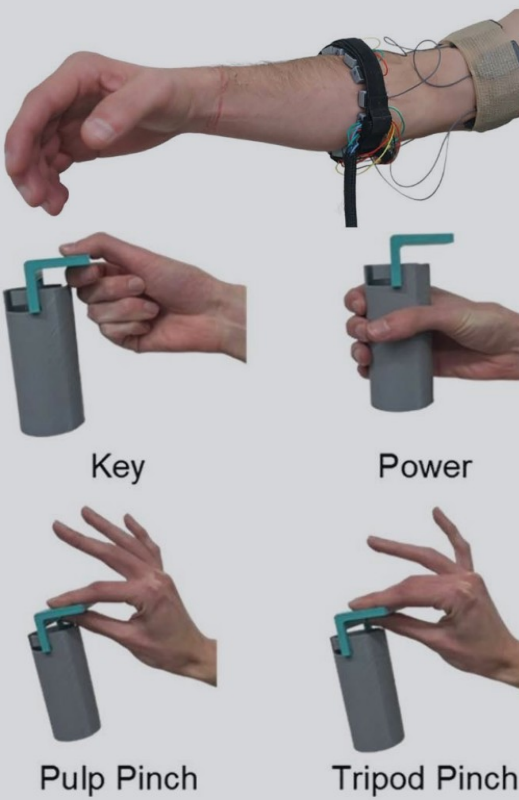
Hand gesture classification is crucial for proper control of human-assistive robotics, such as smart prostheses.²

Ownership of these devices starts with efficient control.



State of the Art

①

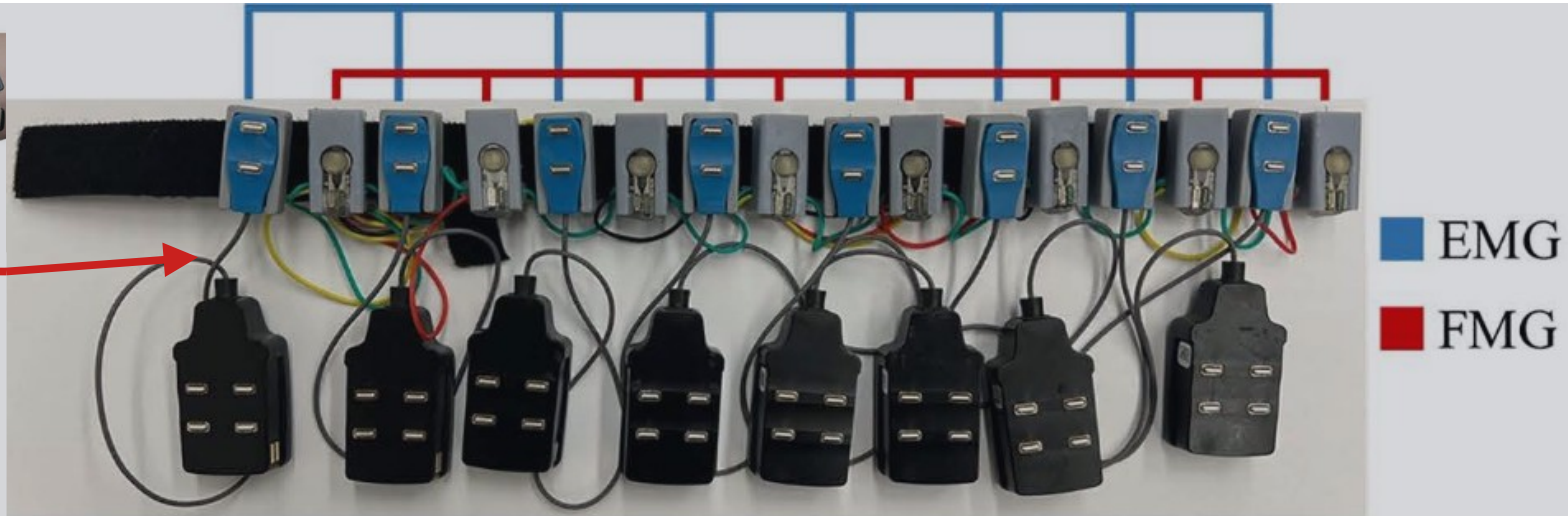


State of the Art

①



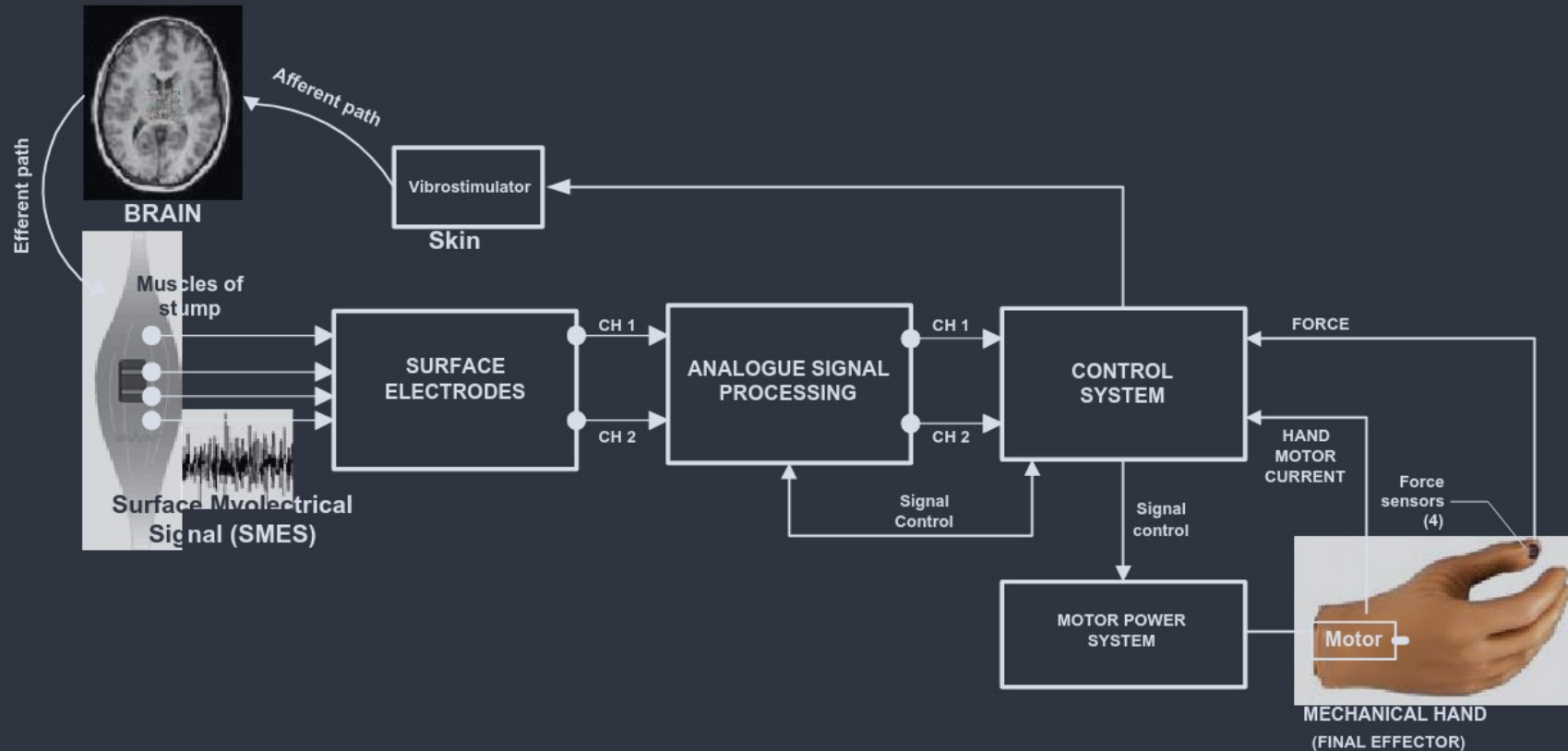
This EMG+FMG
band has 16
sensors!



But can we get similar great results with fewer sensors
to reduce the end cost of the smart prosthetic?

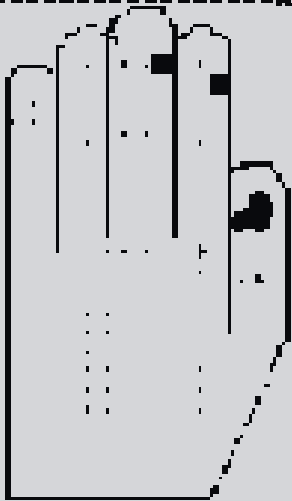
State of the Art

②



State of the Art

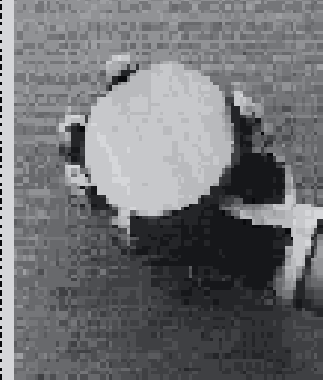
2



a) Cylindrical grasp



b) Spheric grasp



c) Disc grasp

But can we have it detect more gestures?

State of the Art

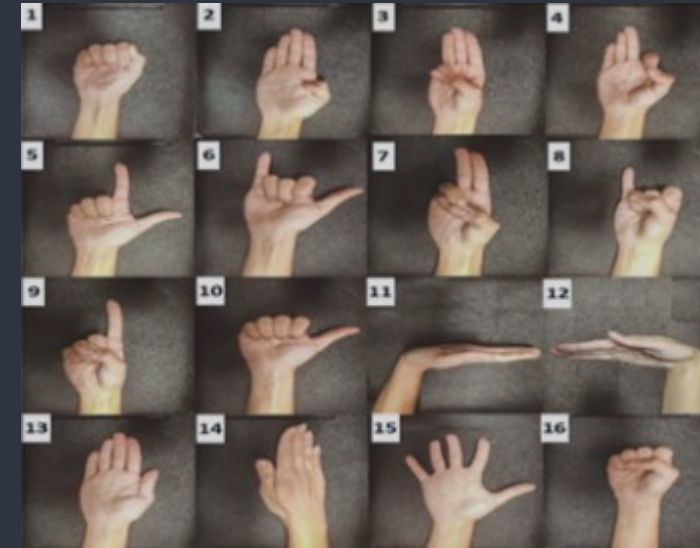
3

(Typical Time Domain Features for EMG)

Feature Name	Description	Formula
Root Squared Zero Order Moment	The feature represents the signal energy. It is normalized by dividing by the median of the zero-order moments from all channels.	$\tilde{m}_0 = \sqrt{\frac{1}{N} \sum_{j=0}^{N-1} x[j]^2}$
Root Squared Second Order Moment	This feature is derived from the second derivative of the signal. It captures spectral properties based on the second derivative of the time-domain signal using the Fourier transform.	$\tilde{m}_2 = \sqrt{\frac{1}{N} \sum_{j=0}^{N-1} (\Delta^2 x[j])^2}$
Root Squared Fourth Order Moment	The feature is derived from the fourth derivative of the signal, capturing higher-order spectral properties from the signal's fourth derivative.	$\tilde{m}_4 = \sqrt{\frac{1}{N} \sum_{j=0}^{N-1} (\Delta^4 x[j])^2}$
Normalized Moments	All moments (zero, second, fourth) are normalized by dividing the moments raised to power k by the median factor λ .	$m_0 = \frac{\tilde{m}_0^k}{\lambda}, \quad m_2 = \frac{\tilde{m}_2^k}{\lambda}, \quad m_4 = \frac{\tilde{m}_4^k}{\lambda}$
Sparseness (S)	This feature quantifies how much energy of a vector is packed into only a few components, measuring sparseness of the signal.	$f_4 = \log(S), \quad S = \left(\frac{m_0}{\sqrt{m_0 - m_2} \sqrt{m_0 - m_4}} \right)$
Irregularity Factor (IF)	As a measure of regularity, this feature captures the ratio of the number of upward zero crossings (ZC) to the number of peaks (NP), using spectral moments.	$f_5 = \log\left(\frac{ZC}{NP}\right) = \log\left(\sqrt{\frac{m_2}{m_0 m_4}}\right)$
Waveform Length Ratio (WL)	WL feature is defined as the ratio of the waveform length of the first derivative to that of the second derivative.	$f_6 = \log\left(\frac{\sum_{i=0}^{N-1} \Delta^2 x_i }{\sum_{i=0}^{N-1} \Delta x_i }\right)$
Coefficient of Variation (COV)	This feature represents the ratio of the standard deviation to the mean of the EMG signal, measuring signal dispersion.	$f_7 = \log(COV) = \log\left(\frac{\sqrt{\frac{1}{N-1} \sum_{j=0}^{N-1} (x[j] - \bar{x})^2}}{\frac{1}{N} \sum_{j=0}^{N-1} x[j]}\right)$
Teager-Kaiser Energy Operator	This non-linear operator measures instantaneous energy changes of signals composed of a single time-varying frequency. The logarithm of the summation of the TKEO is used to represent the EMG energy.	$f_8 = \log(\Psi) = \log\left(\sum_{j=0}^{N-2} x^2[j] - x[j-1]x[j+1]\right)$

(Typical Freq Domain Features for EMG)

Feature	Formula
Energy (E_j)	$E_j = \sum_{i=1}^N W_j(i)^2$
Variance (σ_j^2)	$\sigma_j^2 = \frac{1}{N} \sum_{i=1}^N (W_j(i) - \bar{W}_j)^2$
Standard Deviation (σ_j)	$\sigma_j = \sqrt{\sigma_j^2}$
Waveform Length (WL_j)	$WL_j = \sum_{i=1}^{N-1} W_j(i+1) - W_j(i) $
Entropy (H_j)	$H_j = -\sum_{i=1}^N W_j(i)^2 \log(W_j(i)^2 + c)$



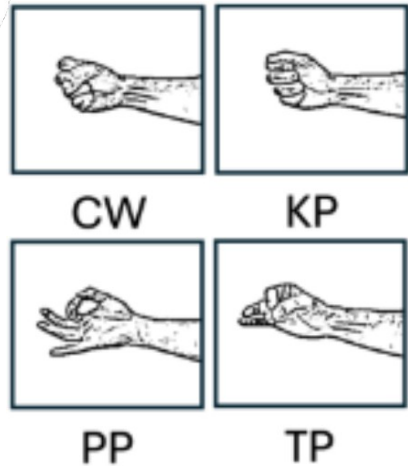
State of the Art



Focus: Construction of an affordable smart prosthetic arm which can perform hand **manipulations** which are useful for purposes in daily life. ✨

State of the Art

Smart Prosthetic Arm

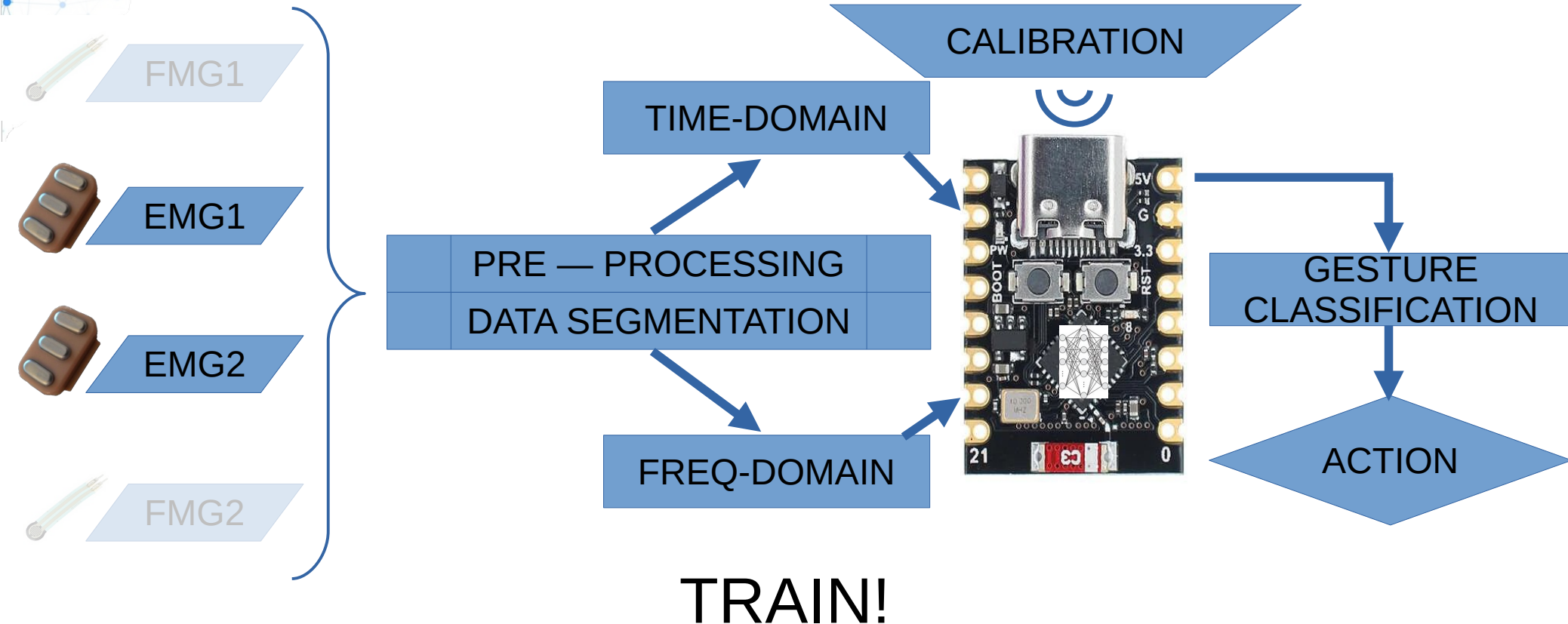


Motor
Intentions

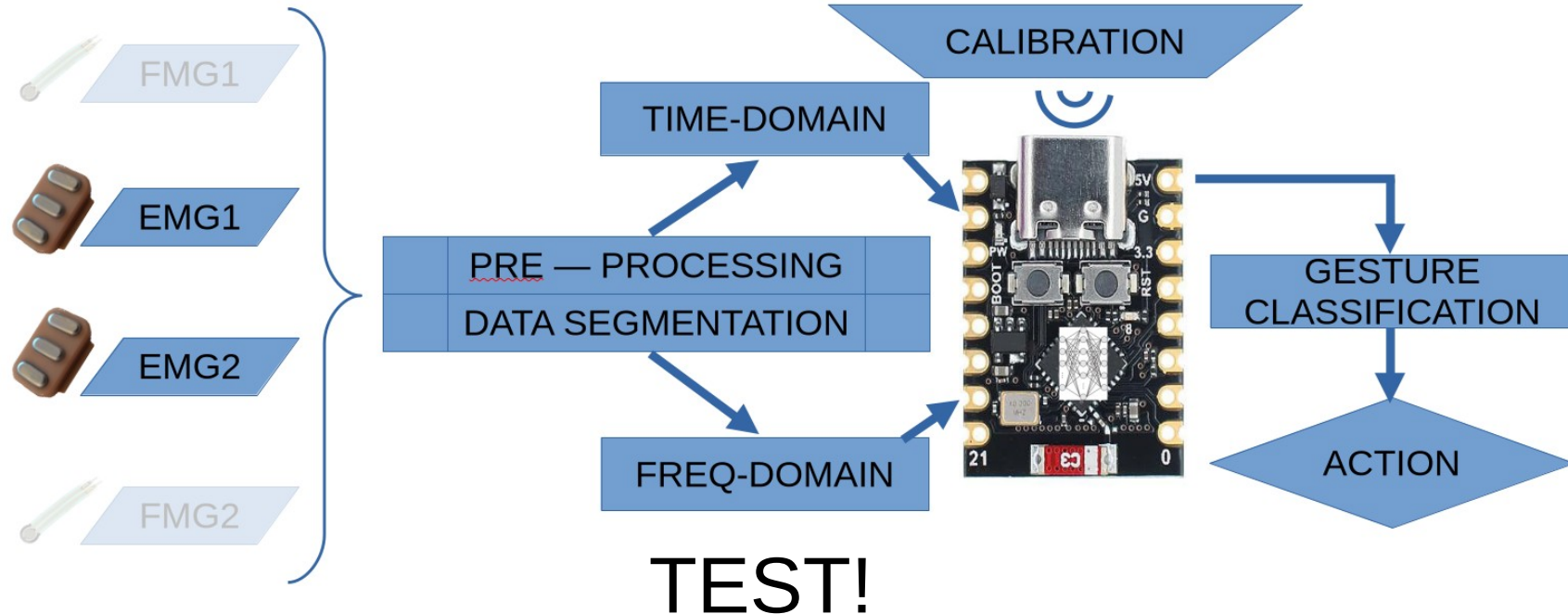


Reliable
Motor
Intention
Predictions

Development Plan

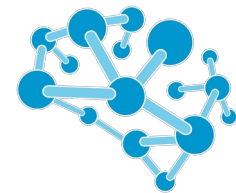


Development Plan



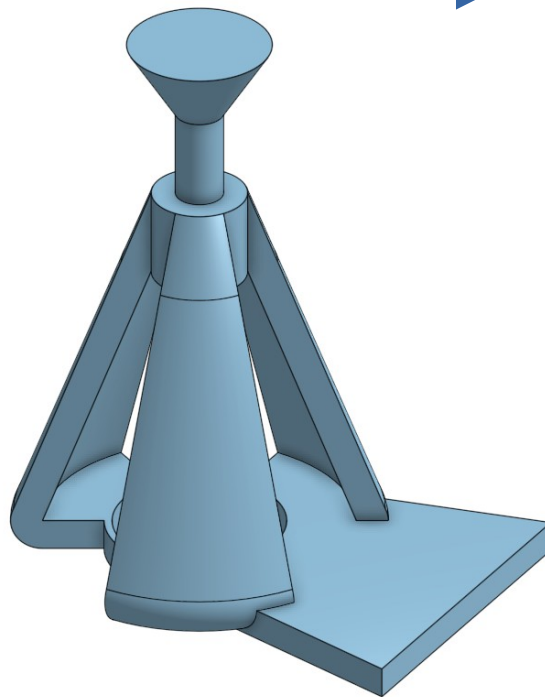


Progress to Date



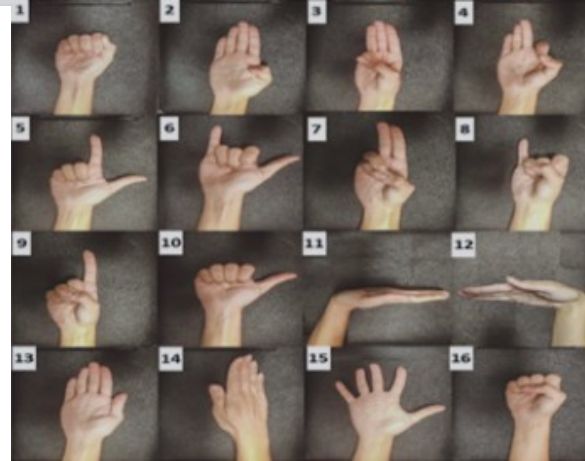
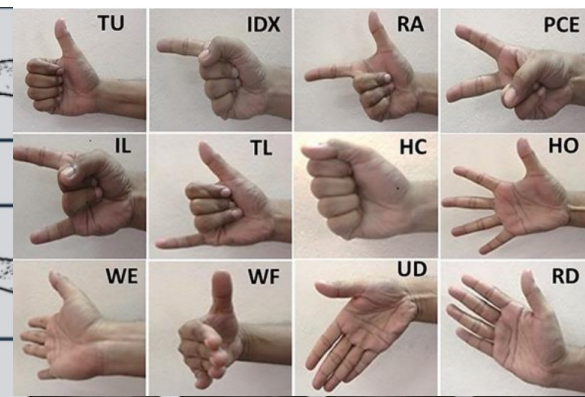
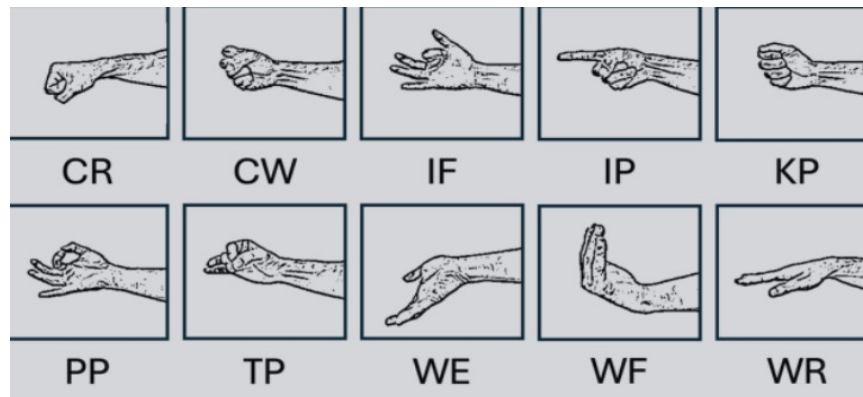
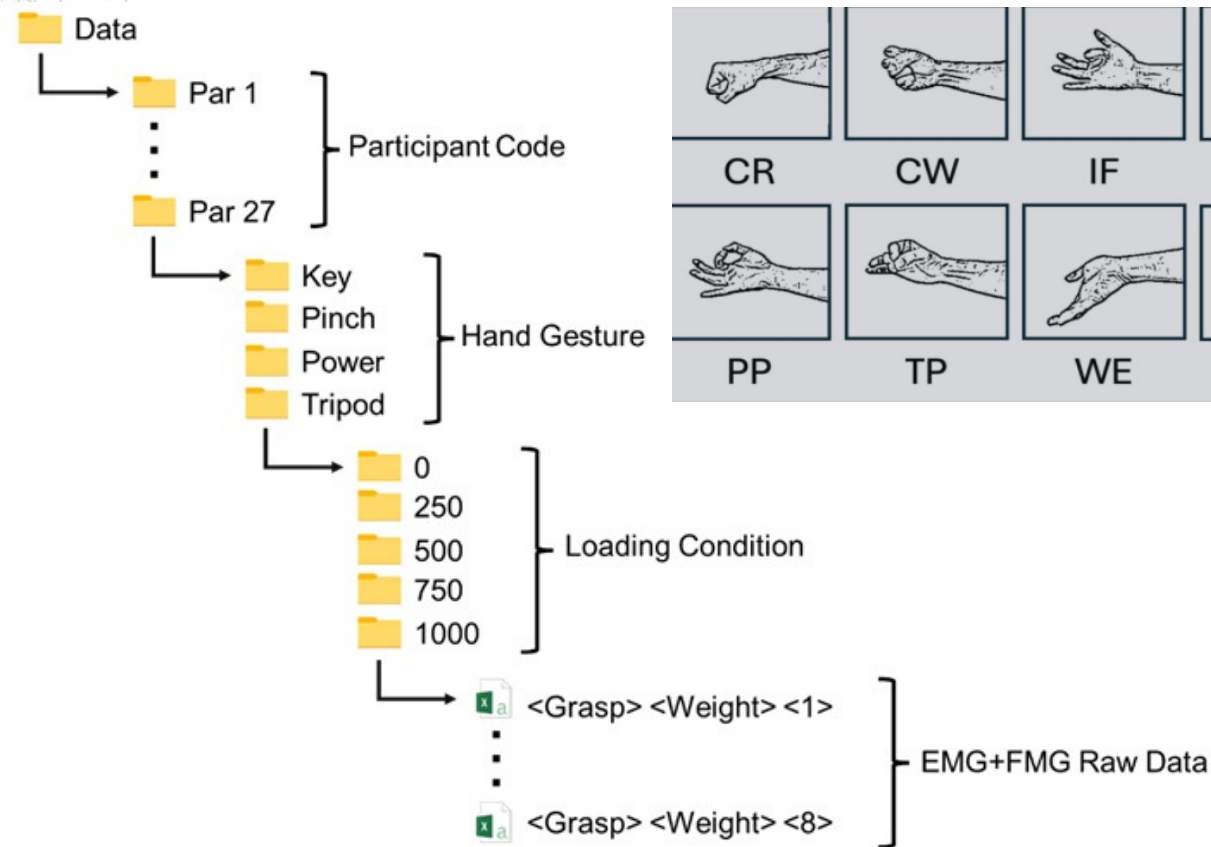
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Used OnShape to make a
manipulandum
(Thank you, Michelle!)



Progress to Date

Found Toy Datasets of sEMG signals to test AI models on



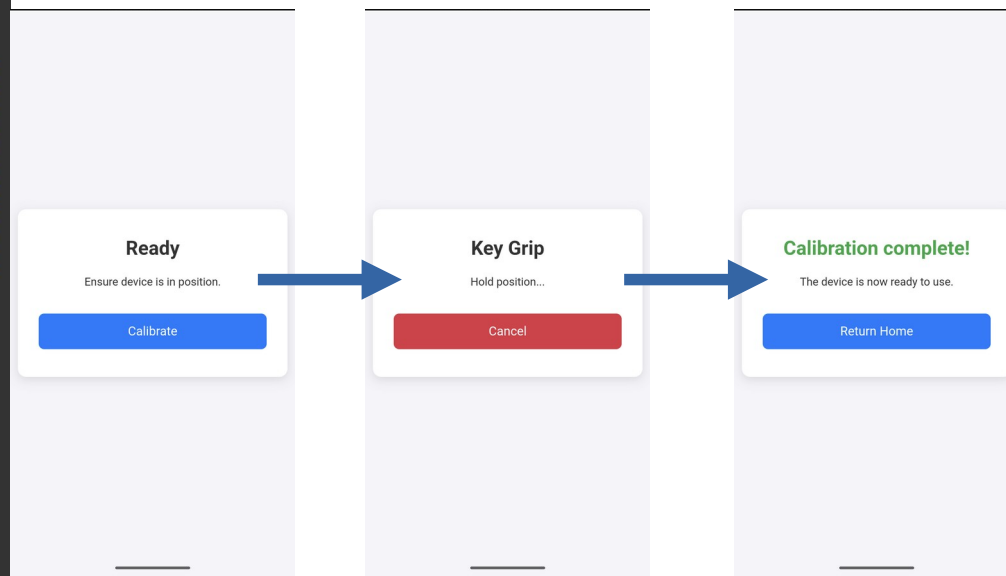
Progress to Date

The screenshot shows the Wokwi web-based IDE. The left pane displays the 'sketch.ino' file with the following code:

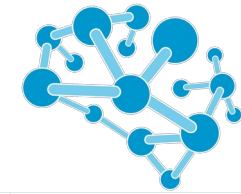
```
1 #include <WiFi.h>
2 #include <WebServer.h>
3
4 // Set your Access Point network credentials
5 const char* ssid = "ESP32_Calibration_App";
6 const char* password = "password123"; // Password must be at least 8 characters
7
8 // Create a web server object that listens on port 80
9 WebServer server(80);
10
11 // =====
12 // HTML, CSS, and JavaScript for the Web App
13 // =====
14 // We use a raw string literal (PROGMEM) to store the HTML efficiently
15 const char index_html[] PROGMEM = R"rawliteral(
16 <!DOCTYPE html>
17 <html lang="en">
18 <head>
19 <meta charset="UTF-8">
20 <meta name="viewport" content="width=device-width, initial-scale=1.0, user-scalable=no">
21 <title>Calibration App</title>
22 <style>
23   body {
24     font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
25     display: flex;
26     justify-content: center;
27     align-items: center;
28     height: 100vh;
29     margin: 0;
30     background-color: #f4f4f9;
31     color: #333;
32     text-align: center;
33   }
34   .container {
35     background: white;
36     padding: 40px 30px;
37     border-radius: 12px;
38     box-shadow: 0 4px 15px rgba(0,0,0,0.1);
39     width: 80%;
40     max-width: 400px;
41   }
42   h2 { margin-top: 0; font-size: 28px; }
43   /* Hide all views by default */
44   .view { display: none; }
45   /* Only show the active view */
46   .active { display: block; }
47
```

The right pane shows a 3D simulation of an ESP32-S3 Sense board with a green play button and a plus icon above it.

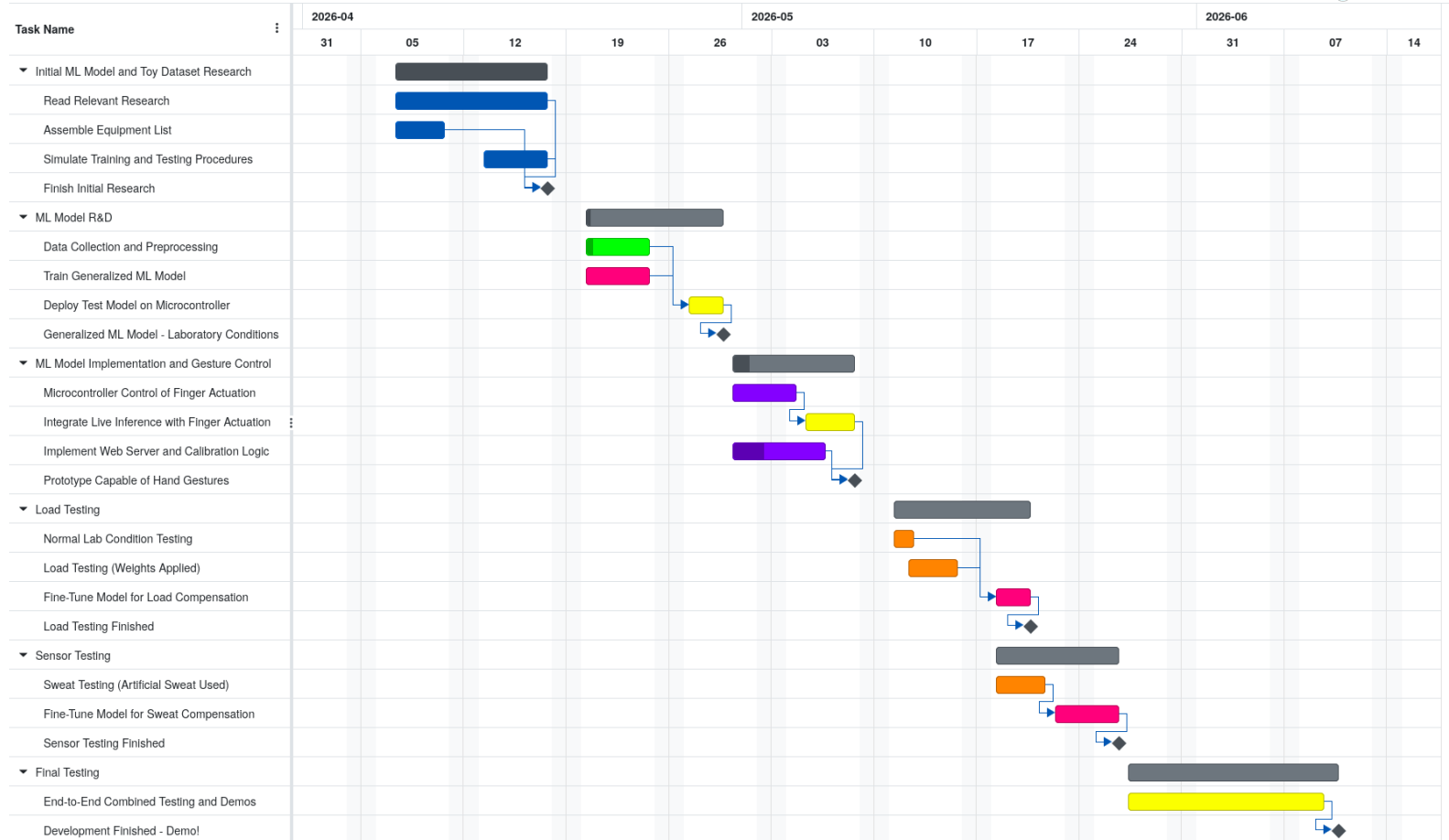
Used Wokwi & ESP32-S3 Sense to simulate working web app that could be used for calibration phase to update weights of ML model.
(Thank you, Daniel!)

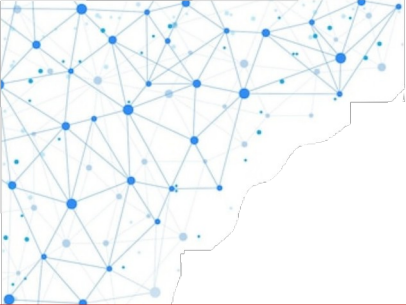


Gantt Diagram



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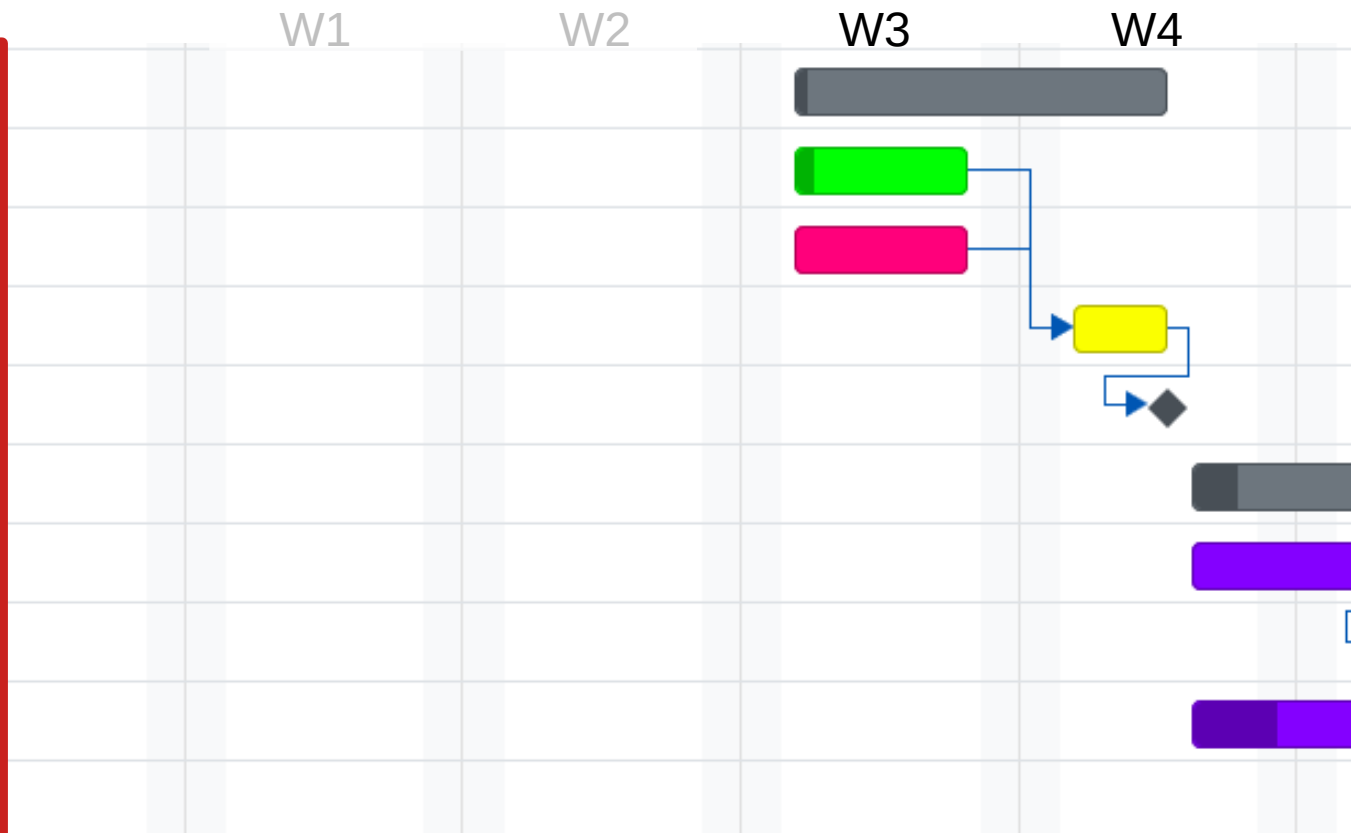




Next Steps



▼ ML Model R&D
Data Collection and Preprocessing
Train Generalized ML Model
Deploy Test Model on Microcontroller
Generalized ML Model - Laboratory Conditions
▼ ML Model Implementation and Gesture Control
Microcontroller Control of Finger Actuation
Integrate Live Inference with Finger Actuation
Implement Web Server and Calibration Logic
Prototype Capable of Hand Gestures



References



- 1) Jaramillo-Yáñez, Andrés, Marco E. Benalcázar, and Elisa Mena-Maldonado. 2020. "Real-Time Hand Gesture Recognition Using Surface Electromyography and Machine Learning: A Systematic Literature Review." *Sensors* 20(9):2467. doi:[10.3390/s20092467](https://doi.org/10.3390/s20092467).
- 2) Young, Peyton R., Kihun Hong, Eden J. Winslow, Giancarlo K. Sagastume, Marcus A. Battraw, Richard S. Whittle, and Jonathon S. Schofield. 2025. "The Effects of Limb Position and Grasped Load on Hand Gesture Classification Using Electromyography, Force Myography, and Their Combination." *PLOS ONE* 20(4):e0321319. doi:[10.1371/journal.pone.0321319](https://doi.org/10.1371/journal.pone.0321319).
- 3) Rios Poveda, Alvaro. 2002. "Myoelectric Prostheses with Sensorial Feedback."
- 4) Aarotale, Parshuram N., and Ajita Rattani. 2024. "Machine Learning-Based sEMG Signal Classification for Hand Gesture Recognition."doi: [10.48550/arXiv.2411.15655](https://doi.org/10.48550/arXiv.2411.15655)
- 5) Hand in bionic hand: The new age of accessible, high-tech prosthetics. (n.d.). The Government of Japan - JapanGov -. Retrieved April 16, 2026, from https://www.japan.go.jp/kizuna/2025/04/high-tech_bionic_prosthetics.html
- 6) RYO™ Smart Bionic Hand – Kawatek. (n.d.). Retrieved April 16, 2026, from <https://kawatek.jp/our-solutions/ryo-smart-bionic-hand/>

*All other images generated with Gemini AI or made by author